

Gradient Descent: The Math of Optimization

Deep Learning

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Contents of This Video

In this video, we will cover:

- The optimization problem in neural networks
- Gradient descent intuition and algorithm
- The learning rate and its effects
- Convergence behavior
- A concrete example with linear regression

The Optimization Problem

Goal: Find weights W and biases b that minimize the loss

$$\min_{W,b} L(W, b) = \min_{W,b} \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i; W, b))$$

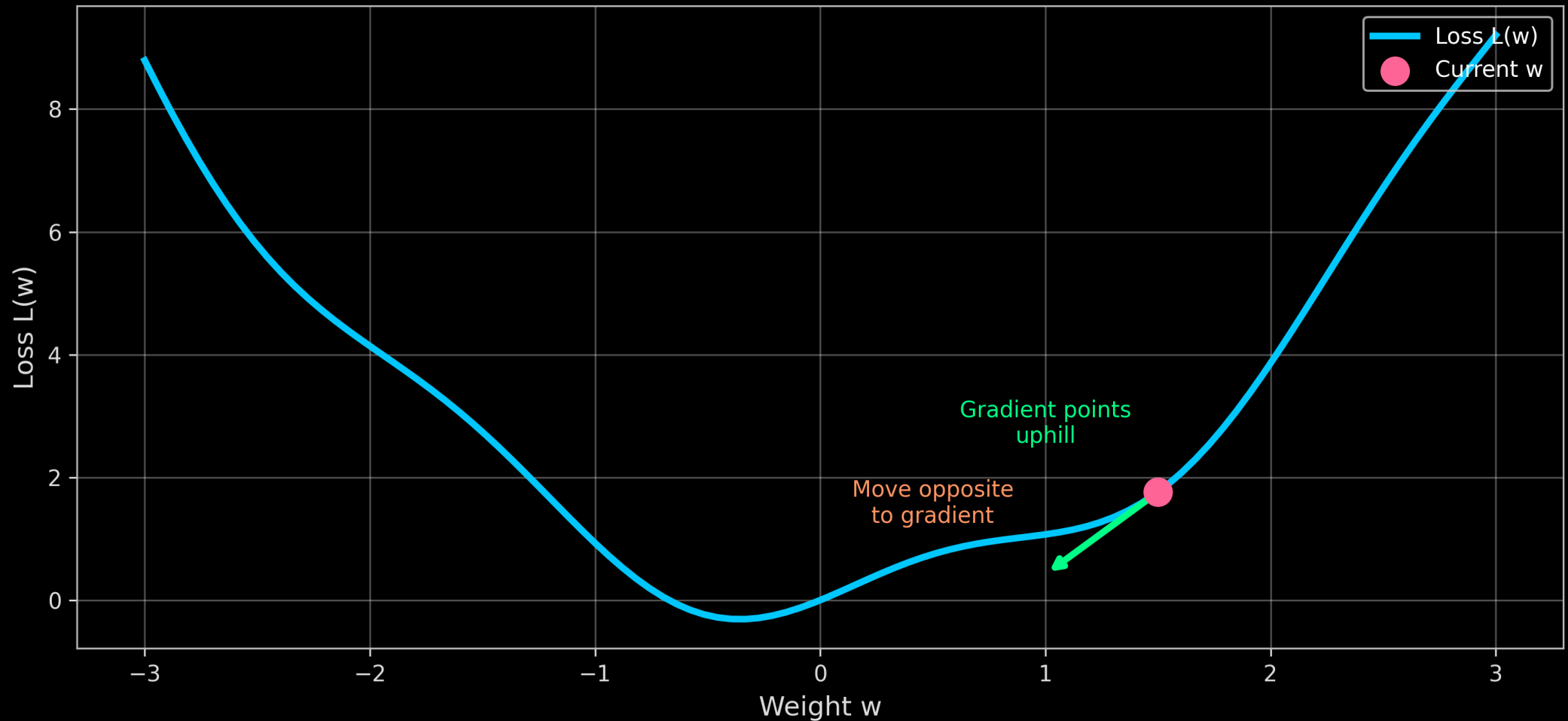
Challenges:

- Cannot solve analytically (non-linear functions)
- Millions of parameters in modern networks
- Loss landscape is complex and high-dimensional

Solution: Iterative optimization with gradient descent

Gradient Descent Intuition

Gradient Descent: Follow the Negative Gradient Downhill



The Gradient Descent Algorithm

Initialize: Random weights w

Repeat until convergence:

$$w \leftarrow w - \alpha \nabla L(w)$$

Where:

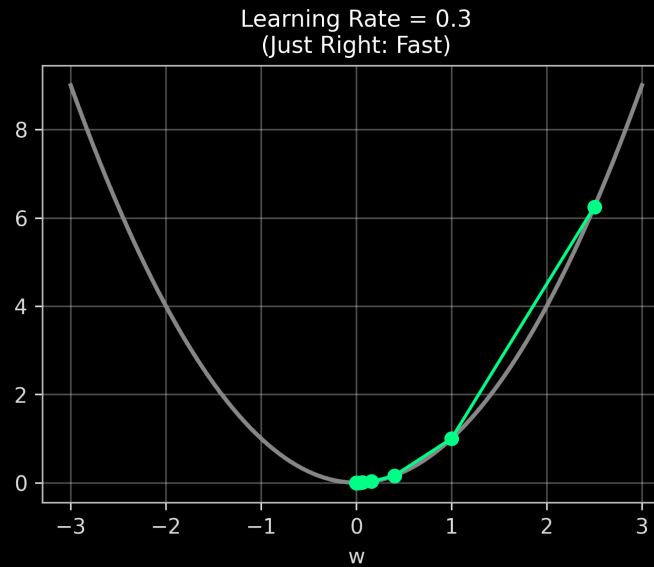
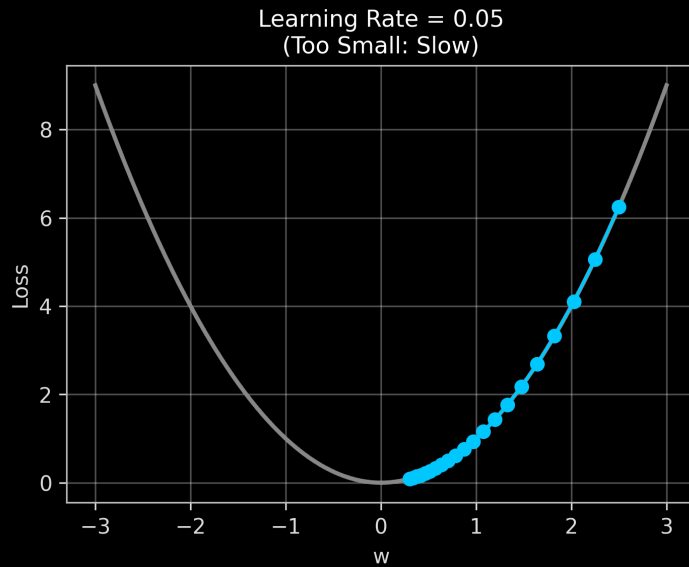
- α is the **learning rate** (step size)
- $\nabla L(w) = \frac{\partial L}{\partial w}$ is the gradient

In neural networks:

$$W^{(l)} \leftarrow W^{(l)} - \alpha \frac{\partial L}{\partial W^{(l)}}$$

$$b^{(l)} \leftarrow b^{(l)} - \alpha \frac{\partial L}{\partial b^{(l)}}$$

Learning Rate: The Critical Hyperparameter



Example: Gradient Descent for Linear Regression

Setup: $\hat{y} = wx$, Loss = MSE

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - wx_i)^2$$

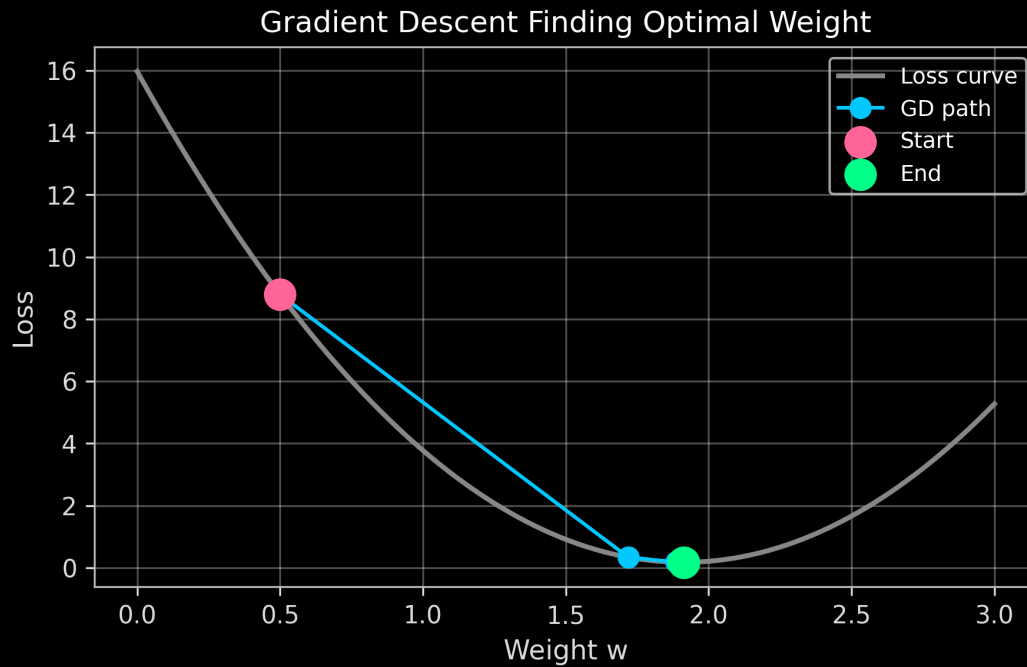
Gradient:

$$\frac{\partial L}{\partial w} = \frac{1}{n} \sum_{i=1}^n -2x_i(y_i - wx_i) = \frac{2}{n} \sum_{i=1}^n x_i(wx_i - y_i)$$

Update:

$$w \leftarrow w - \alpha \cdot \frac{2}{n} \sum_{i=1}^n x_i(wx_i - y_i)$$

Gradient Descent in Action

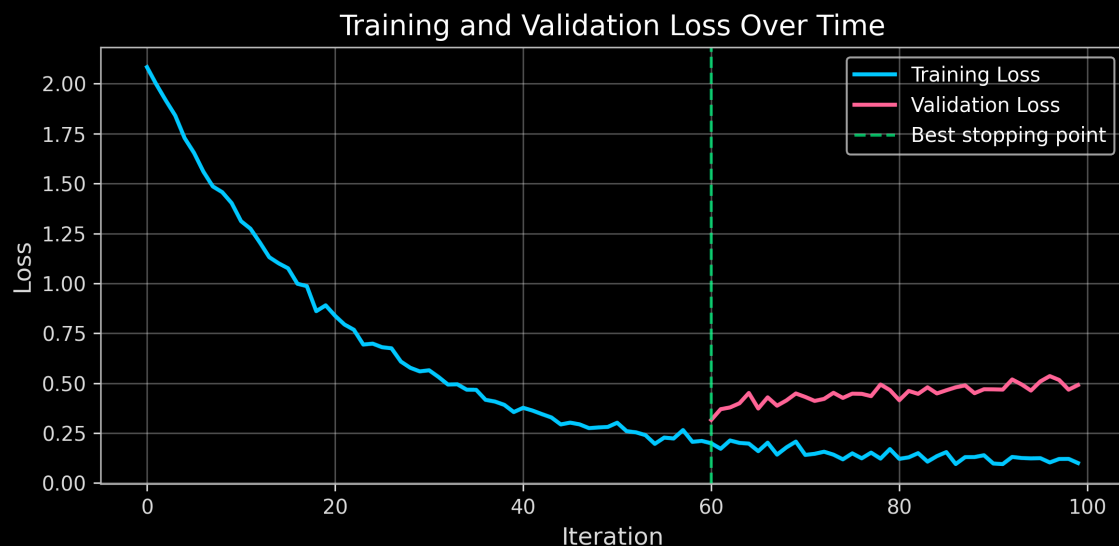


Convergence and Stopping

How do we know when to stop?

- Loss stops decreasing significantly
- Validation loss starts increasing (overfitting)
- Maximum number of iterations reached
- Gradient becomes very small

Typical training curves:



What We've Covered

In this video, we've learned:

- Gradient descent minimizes loss by following the negative gradient
- Update rule: $w \leftarrow w - \alpha \nabla L(w)$
- Learning rate α controls step size (critical hyperparameter)
- Too small: slow convergence; too large: divergence
- Monitor loss curves to determine convergence